



Do Urban Heat Island Effects Reduce Body Mass of Breeding Cooper's Hawks?

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Urbanization and Urban Heat Island (UHI) Effects



Global urban population has grown from 769m in 1950 to 4.4B 2020, or 30% of the human population to 80% [1]



UHI effects estimated to cause $\uparrow$ in ambient temperature by 5°C – 15°C [4]



UHI effects caused by low albedo from excess impermeable surfaces, air conditioners usage and vehicle and overall CO_2 emissions [3]

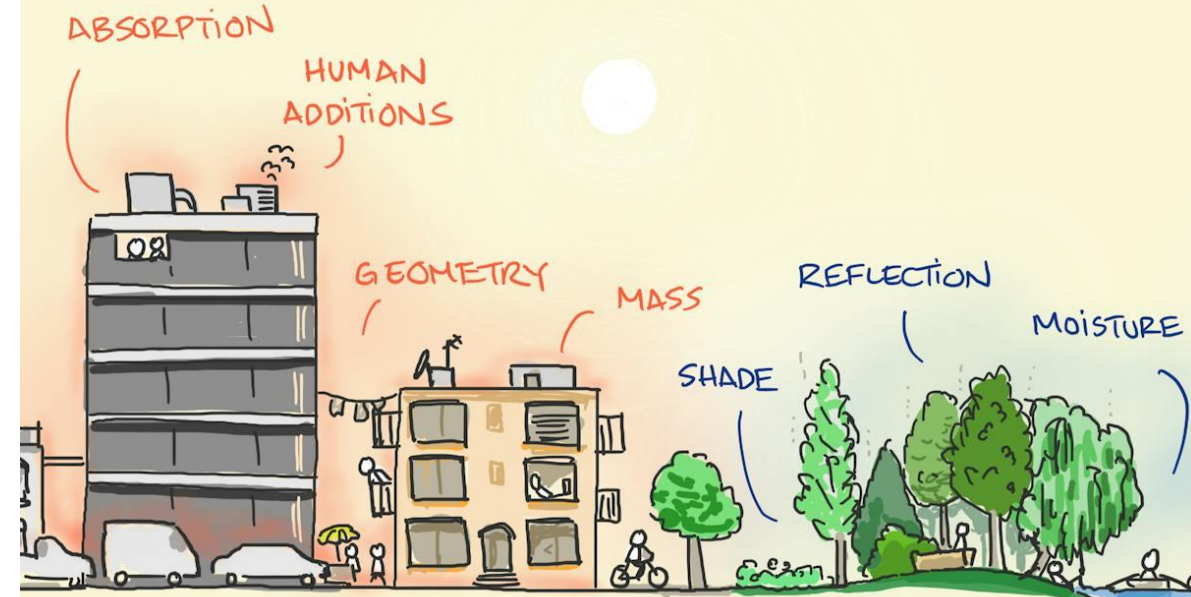


UHI effects also include $\downarrow$ in vegetation and evapotranspiration, $\downarrow$ air quality, $\uparrow$ in particulates and pollutants in the air (SO_2 , NO_x) [5]



$\uparrow$ Human presence/activity = $\uparrow$ urban heat island (UHI) effects [2]

HEAT ISLANDS



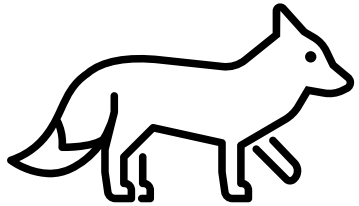
URBAN AREAS AND STRUCTURES ABSORB AND RETAIN HEAT

MAKING THEM HOTTER THAN SURROUNDING NATURAL AREAS

sketchplanations

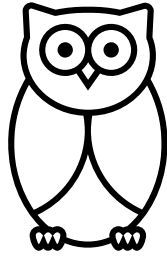
Synurbization

-Wildlife Adaptations to Urban Environments^{[7][8]}



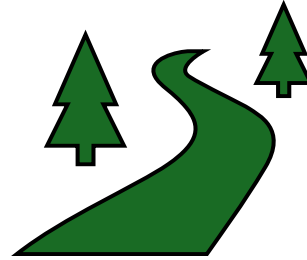
Behavioral changes:

- ↑ population density
- ↑ boldness
- ↑ intra-specific aggression
- Changes in dwelling



Changes in physiology:

- Immunity system changes
- Changes in body mass



Urbanization is directly correlated to ↓ **in biodiversity**
↑ **in habitat homogeneity**



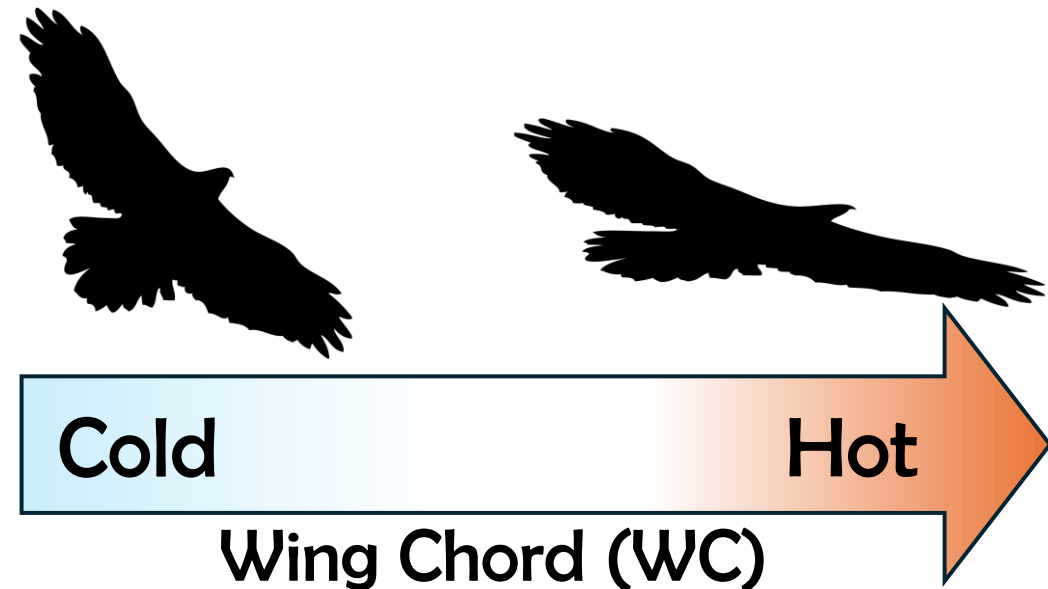
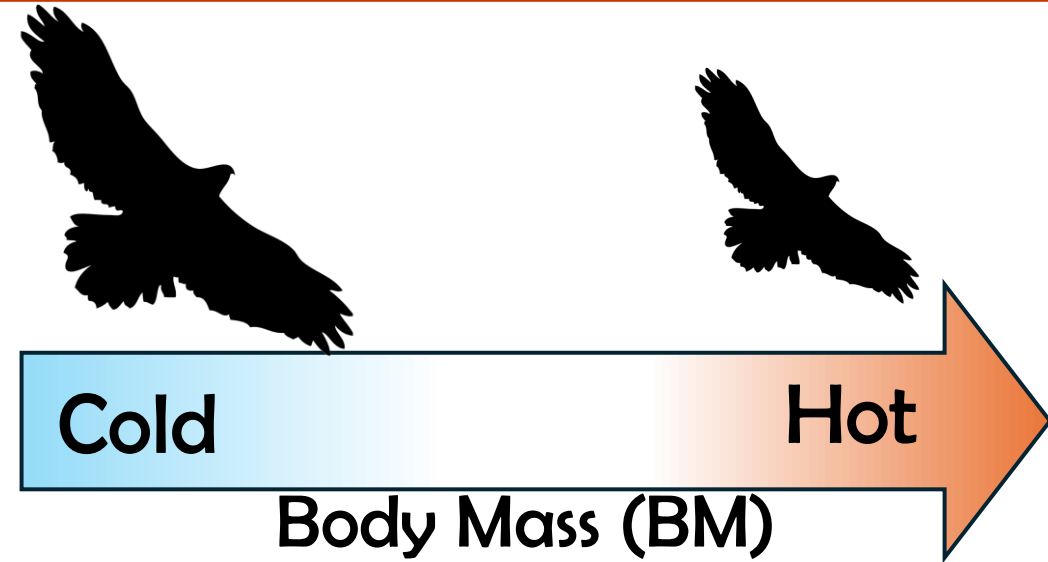
Cooper's Hawks (*Astrus cooperii*)

- Crow-sized, avivorous raptor
- Females are ~1.7x larger than males (highest in all NA raptors) ^[9]
- Exhibits breeding and nonbreeding resident sites as well as permanent resident sites ^[10]
- High site fidelity
- Single brooded
- Pioneers large metropolitan environments at rapid rates ^[11]
- Exhibits high inter-population morphometric plasticity ^[12]



Body Mass (BM) and Wing Chord (WC) Metrics

- $\uparrow$ in breeding ground temperatures for avians are significantly correlated towards $\downarrow$ **BM** ^[13]
- $\uparrow$ of ambient temperatures in breeding and non-breeding ranges related to $\uparrow$ **WC** length ^[13]
 - $\uparrow$ in WC length can be due to predator avoidance ^[14], foraging behavior ^[15] and flight efficiency ^[16]
- $\downarrow$ in BM due to $\uparrow$ heat island effects is also observed in ground beetles and rodents ^{[17][18][19]}

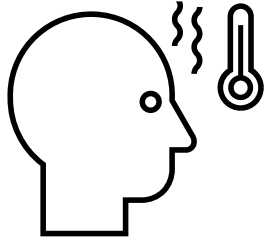


Bergmann's and Allen's Rule

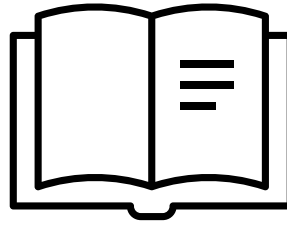
- Populations in warmer climates tend to be smaller than those in colder climates^[20]
- Homeotherms in warmer climates have longer appendages than those in colder climates^[21]
- Increases surface area to volume ratio which aids in heat dissipation



Why Investigate?



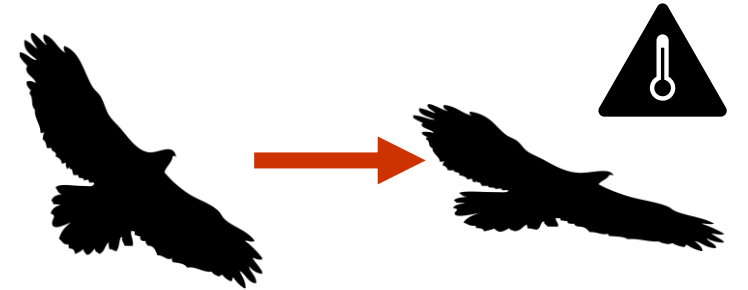
Birds have limited room for internal overheating before heat prostrations = $\uparrow$ in adaptation pressure ^[18]



Literature

knowledge gaps on urban-heat-island effects in wildlife

Zero studies on morphometric changes among raptor populations in relation to warming thermal environments



Given Bergmann's Rule, Allen's Rule, and Weeks et al. (2019)^[12], we would expect smaller body masses and longer wing chords among bird populations in warmer areas

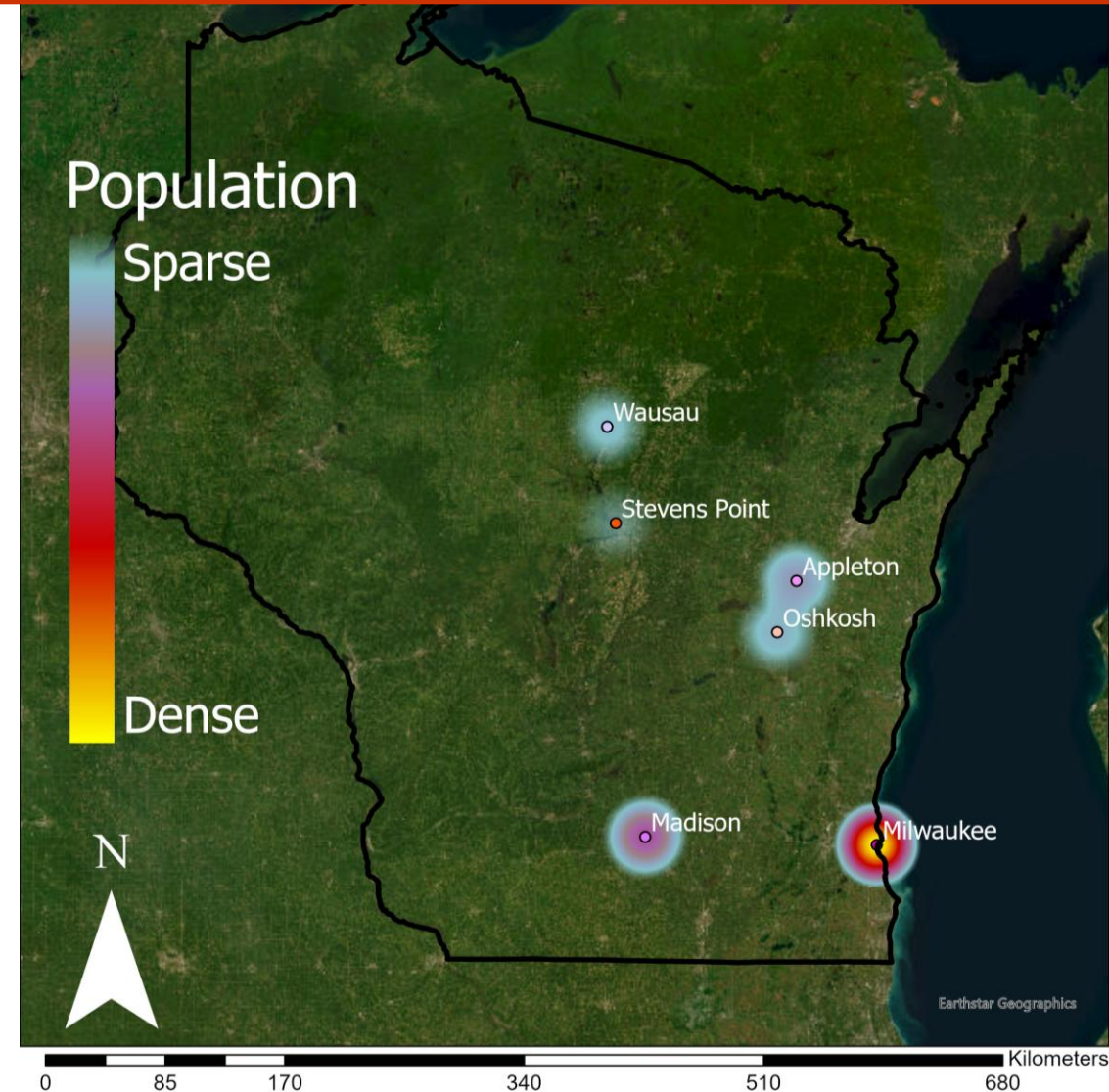
Methods

- We trapped and individually marked **$n = 178$** adult breeding Cooper's Hawks from **2011 to 2024** during **mid-June to mid-July**
- We chose nests based of hawks with existing USGS BBL bands
- We deployed a mist net near each nest with a live decoy Great Horned Owl
- We took standard morphological data, including **WC** and **BM**
- Six study sites:
 - **Appleton:** $n = 2$ females, $n = 2$ males
 - **Madison:** $n = 7$ females, $n = 12$ males
 - **Milwaukee:** $n = 18$ females, $n = 30$ males
 - **Oshkosh:** $n = 22$ females, $n = 30$ males
 - **Stevens Point:** $n = 27$ females, $n = 24$ males
 - **Wausau:** $n = 2$ females, $n = 2$ males



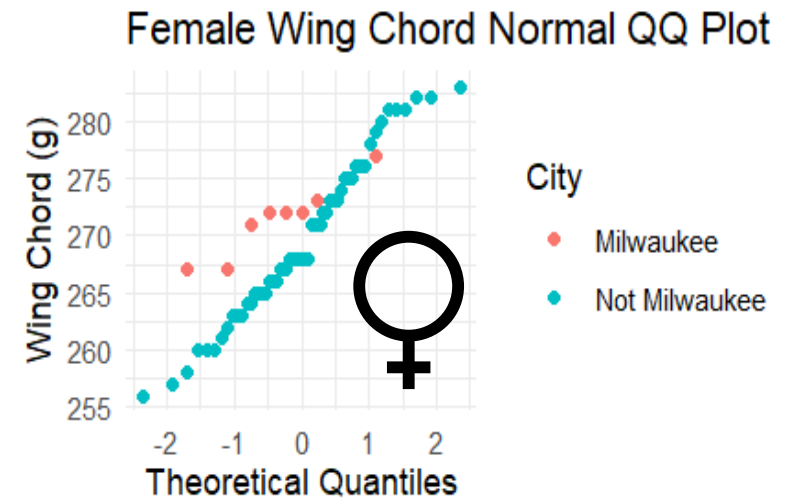
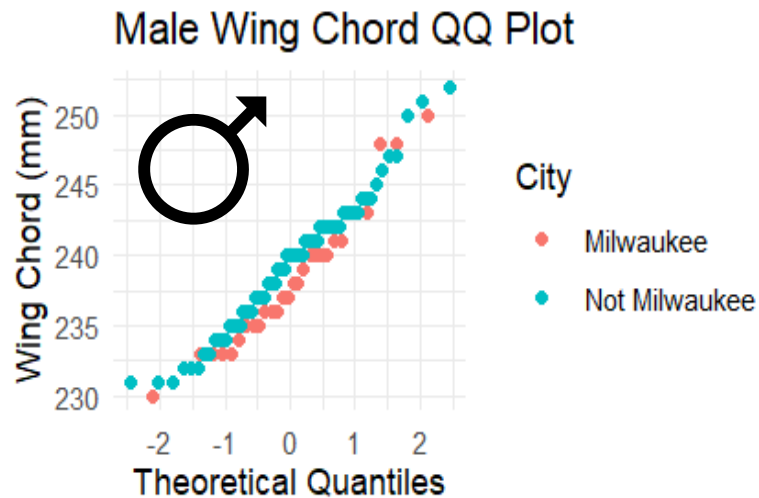
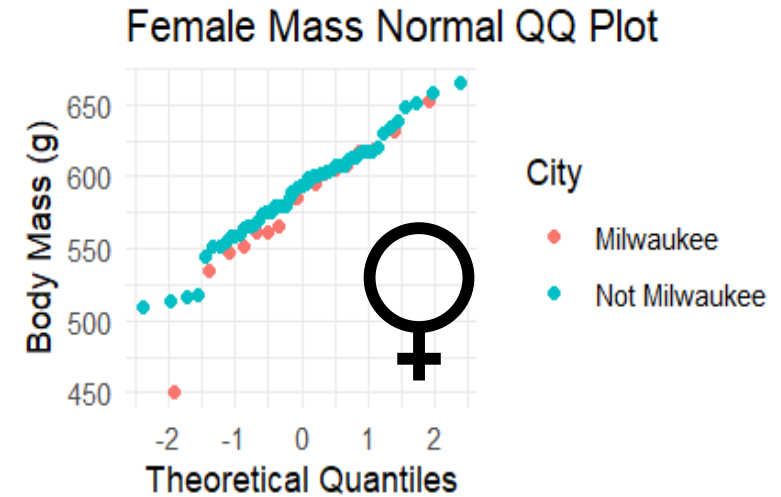
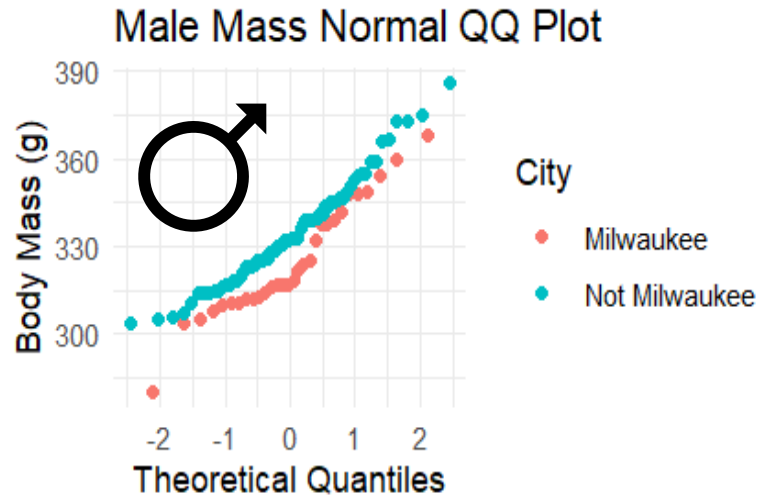
Milwaukee vs xMilwaukee

- Milwaukee's human population >1,400,000 as of 2025 ^[21]
 - Madison = 519,000
 - Oshkosh = 171,730
 - Appleton = 259,442
 - Stevens Point area = 70,733
 - Wausau area = 138,000
- Wei et al. 2024 suggests once a population hits 1,000,000, UHIs start to increase 10-fold, starting at 8° C. ^[21]
- All non-Milwaukee cities were pooled together for this reason



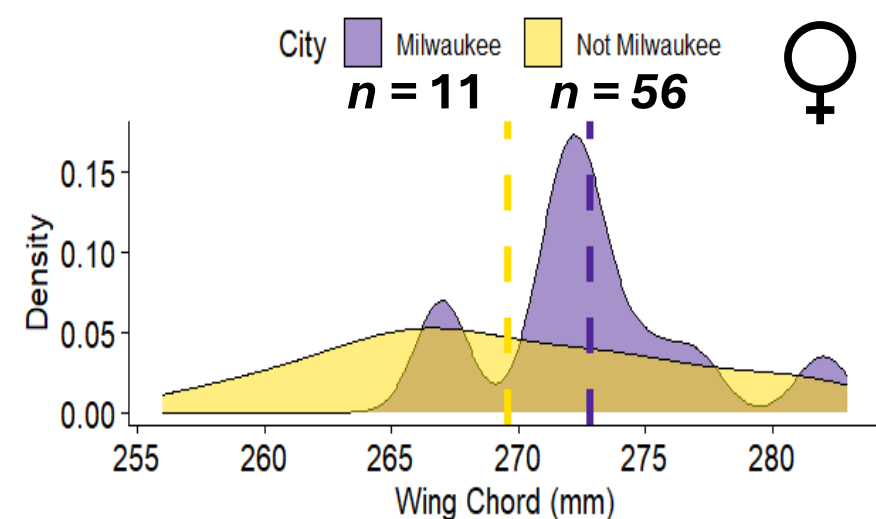
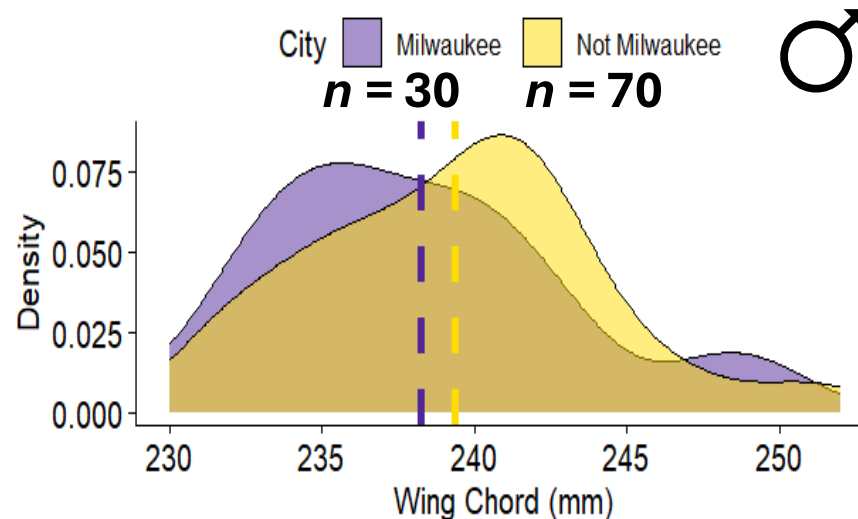
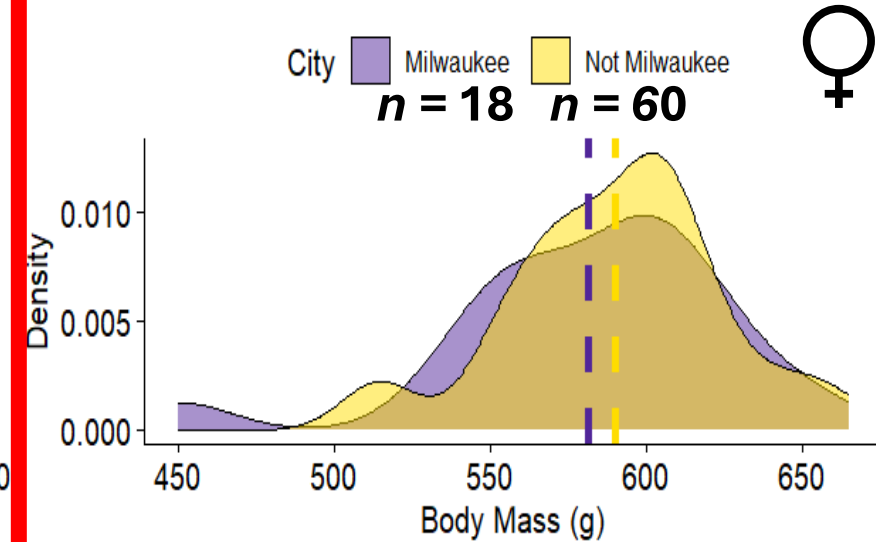
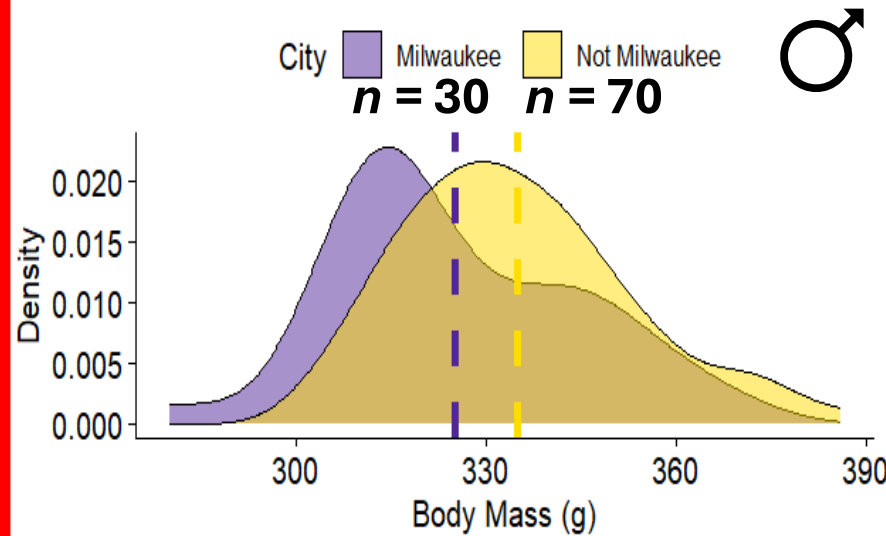
Analysis

- Shapiro-Wilk for normalcy
 - All four datasets are normally distributed
- Unpaired Student's T-Test to determine significant differences among mass and wing chord metrics



Results

- There was a **significant difference in male body mass**
- $t(98) = -2.43, p = 0.017$
- Milwaukee male body masses (g)
 - Mean = 325.1 g
 - SD = 19.6
- xMilwaukee male body masses (g)
 - Mean = 334.9 g
 - SD = 18.1



Discussion

- Novel discovery in the areas of Synurbization, eco-morphology and climate change
- First study to provide empirical evidence towards raptor morphological variation amongst warmer thermal environments
- Males Cooper's hawks have high energetic demands during mid-June to mid-July
 - Constant foraging
 - Highest extra-pair copulation rate of NA raptors (linked with urban environments due to higher territory density) ^[22]
 - Nest provisioning
 - Nest defense
- No direct linkage of urban heat island effects causing morphological variation



Management Implications

- Climate change exacerbates UHI effects at a significantly higher rate than non-urban environments^[4]
- Properly planned urban green spaces mitigate UHI effects ^[24]
- Ease of access to shade and water can ease overheating risks



Future Research

- Any other wildlife species with similar latitudinally arranged urban and x-urban populations
- Cooper's hawk behavioral study to analyze overheating symptoms
- Comparative study of Milwaukee's UHI effects on wildlife relative to other urban environments
 - Lake Michigan provides a cooling effect on Milwaukee



Acknowledgements

- Advisor: Dr. Robert Rosenfield
- UWSP Department of Biology





Questions?

Supplemental Summary Table

Sex	City	Metric	n	mean	median	min	max	q1	q3	IQR	MAD	sd	se	ci
M	Milwaukee	Mass (g)	30	325.1	317.5	280	368	312	338.8	26.8	12.6	19.6	3.6	7.3
	Not													
M	Milwaukee	Mass (g)	70	334.9	332.5	304	386	323	345	22	17.8	18.1	2.2	4.3
M	Milwaukee	Wing Chord (mm)	30	238.2	237.5	230	250	235	240.8	5.8	4.4	4.9	0.9	1.8
	Not													
M	Milwaukee	Wing Chord (mm)	70	239.4	240	231	252	236	242	6	4.4	4.7	0.6	1.1
F	Milwaukee	Mass (g)	18	581.4	590	450	653	561.2	607.2	46	40.8	54.4	10.7	22.6
	Not													
F	Milwaukee	Mass (g)	60	590.1	594	510	665	569.2	608.2	39	28.9	33.9	4.4	8.7
F	Milwaukee	Wing Chord (mm)	11	272.8	272	267	282	271.5	274	2.5	1.5	4.2	1.3	2.8
	Not													
F	Milwaukee	Wing Chord (mm)	56	268.8	268	225	283	264.8	275	10.2	7.4	9.2	1.2	2.5

Supplemental Shaprio-Wilk Results

Table 1. Shapiro-Wilk test results for normality taken amongst different datasets of measurements of Wisconsin Cooper's hawks. Datasets are grouped by city and sex.

Variable	Sex	City	W Statistic	p value
Mass	Male	Milwaukee	0.947	0.143
WingChord	Male	Milwaukee	0.948	0.148
Mass	Male	Not Milwaukee	0.968	0.075
WingChord	Male	Not Milwaukee	0.973	0.141
Mass	Female	Milwaukee	0.914	0.102
WingChord	Female	Milwaukee	0.917	0.292
Mass	Female	Not Milwaukee	0.978	0.333
WingChord	Female	Not Milwaukee	0.860	0.000

Supplemental S. T-Test Results

Student T-Test Results for comparing mass and wing chord of Cooper's Hawks found in Milwaukee to five other developed Wisconsin cities

Sex	Dependent Variable	t-statistic	df	p-value	Effect Size	CI_lower	CI_upper
Male	Mass	-2.423	98	0.017*	-0.529	-0.961	-0.093
Male	WingChord	-1.077	98	0.284	-0.235	-0.663	0.195
Female	Mass	-0.885	76	0.379	-0.238	-0.765	0.291
Female	WingChord	1.469	64	0.147	0.485	-0.170	1.136

* = p-value is less than the alpha of 0.05

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